

Figure 1-28. Watchdog Counter Register (WDCNTR)

15	8	7	0
Reserved		WDCNTR	
R-0		R-0	

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 1-28. Watchdog Counter Register (WDCNTR) Field Descriptions

Bits	Field	Description
15-8	Reserved	Reserved
7-0	WDCNTR	These bits contain the current value of the WD counter. The 8-bit counter continually increments at the watchdog clock (WDCLK), rate. If the counter overflows, then the watchdog initiates a reset. If the WDKEY register is written with a valid combination, then the counter is reset to zero. The watchdog clock rate is configured in the WDCR register.

Figure 1-29. Watchdog Reset Key Register (WDKEY)

15	8	7	0
Reserved		WDKEY	
R-0		R/W-0	

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 1-29. Watchdog Reset Key Register (WDKEY) Field Descriptions

Bits	Field	Value	Description ⁽¹⁾
15-8	Reserved		Reserved
7-0	WDKEY	0x55 + 0xAA Other value	Refer to Table 1-26 for examples of different WDKEY write sequences. Writing 0x55 followed by 0xAA to WDKEY causes the WDCNTR bits to be cleared. Writing any value other than 0x55 or 0xAA causes no action to be generated. If any value other than 0xAA is written after 0x55, then the sequence must restart with 0x55. Reads from WDKEY return the value of the WDCR register.

⁽¹⁾ This register is EALLOW protected. See [Section 1.5.2](#) for more information.

Figure 1-30. Watchdog Control Register (WDCR)

15															8																										
Reserved																																									
7							6							5							3							2							0						
WDFLAG							WDDIS							WDCHK							WDPS																				
R/W1C-0							R/W-0							R/W-0							R/W-0																				

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 1-30. Watchdog Control Register (WDCR) Field Descriptions

Bits	Field	Value	Description ⁽¹⁾
15-8	Reserved		Reserved
7	WDFLAG	0 1	Watchdog reset status flag bit The reset was caused either by the $\overline{XRS}$ pin or because of power-up. The bit remains latched until you write a 1 to clear the condition. Writes of 0 are ignored. Indicates a watchdog reset ($\overline{WDRST}$) generated the reset condition. .
6	WDDIS	0 1	Watchdog disable. On reset, the watchdog module is enabled. Enables the watchdog module. WDDIS can be modified only if the WDOVERRIDE bit in the SCSR register is set to 1. (default) Disables the watchdog module.

⁽¹⁾ This register is EALLOW protected. See [Section 1.5.2](#) for more information.